Course Title:	Probability and Statistics for Data Science and Cyber Security Engineers	Prerequisite: NA	No. of hrs./sem.					Total no. of hrs./sem.	Total no. of Credits* (= Total no. of hrs. per sem. / 30)
		Relevant SDG	L	T	P	TW# SL	TW# MP		
Course Code:	25DC210	9 & 12	45	0	30	20	25	120	4

Relevant PEO(s):	**PEO1:** ***Innovate*** scalable solutions by leveraging AI and Data Science expertise.
Relevant PSO(s):	**PSO1:** ***Evaluate*** the performance and applicability of intelligent systems across diverse domains
Course Competency:	***Use*** *the principles of Probability and Statistics to solve problems in Data Science and Cyber Security.*
Course Category:	**BSC** – Basic Science Course

Course Outcomes: *At the end of the course, students will be able to:*

CO1	***Evaluate*** the decisions of intelligent systems that uses probability.
CO2	***Model*** the data that has uncertainties.
CO3	***Predict*** the network traffic having uncertainties**.**
CO4	***Justify*** Artificial Intelligence decisions statistically**.**
CO5	***Validate*** the data driven models using statistical tests.

Learning Unit: I	**Probability**	**15 Hours**

Probability - Definitions and axioms-independent events- conditional probability- Total Theorem - Baye's theorem- Problems-Naive Bayes Algorithm in Machine Learning - Bayesian probability in risk assessment and decision-making under uncertainty.

Learning Unit: II	**Random Variables**	**15 Hours**

Discrete and Continuous Random Variables-Probability Density Function-Cumulative Distribution Function. Two-Dimensional Random Variables: Discrete and Continuous Random variables, Marginal Probability Distribution-Conditional Probability Distribution-Independent Random Variables - Random variables and their distributions in data science and cyber security for analyzing data patterns and detecting anomalies.

Learning Unit: III	**Probability Distributions**	**15 Hours**

Probability distributions. Discrete: Binomial distribution and Poisson distribution. Continuous: Normal distribution, Exponential distribution–Probability distributions in data science and cyber security for modelling events, predicting outcomes, and assessing risks.

Learning Unit: IV	**Statistics**	**15 Hours**

Measures of Central tendency-Time Series Components of Time Series -Measurement of Trend-Freehand or Graphic Method-Moving Average Method-Correlation, Rank Correlation and Regression – Statistical measures for pattern identification, anomaly detection, and data-driven decision-making in Data Science and Cyber Security systems.

Learning Unit: V	**Testing of Hypothesis**	**15 Hours**

Test for single proportion, difference of proportions, Tests for single mean, difference of means. - Confidence intervals of parameters - Student t-test, F-test and Chi-square test for goodness of fit and independence of attributes –Hypothesis tests and confidence intervals for assessing risks and supporting decision-making in Data Science and Cyber Security.

Total	**75 Hours**

*# **TW** – Self Learning (SL), and Micro Project (MP) hours are indicative of learning hours spent by students beyond regular working hours.*

Suggested Learning Resources

S. No.	Title of Book	Author	Publication details including ISBN
1	Higher Engineering Mathematics	B.S. Grewal.	Khanna Publishers, 43rd Edition,2015, ISBN:9788174091955
2	Higher Engineering Mathematics	B. V. Ramana	Tata McGraw Hill New Delhi, 29th Reprint,2017, ISBN: 9780070634190
3	Probability, Statistics and Random Processes	T. Veerarajan,	3rd Edition, Tata McGraw-Hill, New Delhi, 2017.
4	Advanced Engineering Mathematics	Erwin Kreyszig	10thEdition, John Wiley & Sons, 2017
5	Fundamentals of Mathematical Statistics	S.C. Gupta, V.K. Kapoor	Sultan Chand &Sons, 12th Revised Edition 2020.
6	Introduction to Probability Theory	P. G. Hoel, S. C. Port and C. J. Stone	Universal Book Stall, 2003(Reprint).
Course Developed by:		Dr. K. Rebecca Jebaseeli Edna, Associate Professor, Division of Data Science and Cyber Security.	
Reviewed by:		Dr. A. Hepzibah Christinal, Professor, CSE.	
Recommended by the Board of Studies on:		30th January 2026	
Approved by the Academic Council on:			

** **30** learning hours will be counted as **One Credit** as per Sec. 3.2.4, pp. 35, UGC-NCrF April 2023; as per Sec. 4.3, pp.45 of UGC-NHEQF May 2023 and as per Table no. 1.2.2.1, Criterion 1.2.2, pp. 10 of NBA SAR August 2024.*